Mathematics A

Unit 2 • Chapter OL-T55 Classified

Cross-session • chapter-organised candidate

22 classified items • 79 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

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Contents

Unit 2 • Unit 2 • Chapter OL-T55 • 22 items • 79 marks

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Cumulative Frequency Diagrams

Unit 2 • 79 marks • 22 item(s)

June 2025 | 4MA1-2H Q13 | 6 marks
Plot a cumulative-frequency curve from class frequencies

June 2024 | 4MA1-2H Q13 | 7 marks
Estimate the number above or between thresholds

November 2023 | 4MA1-1H Q13 | 1 marks
Estimate the median from a cumulative-frequency diagram

November 2023 | 4MA1-1H Q13 | 3 marks
Estimate a percentage from a cumulative curve

June 2023 | 4MA1-1H Q12 | 4 marks
Estimate the median from a cumulative-frequency diagram

November 2024 | 4MA1-2H Q13 | 6 marks
Estimate the number above or between thresholds

June 2022 | 4MA1-2H Q13 | 7 marks
Estimate the number above or between thresholds

January 2022 | 4MA1-1H Q12 | 6 marks
Estimate a percentage from a cumulative curve

January 2020 | 4MA1-1H Q12 | 2 marks
Estimate IQR from a cumulative-frequency diagram

January 2020 | 4MA1-1H Q12 | 3 marks
Estimate a percentage from a cumulative curve

January 2021 | 4MA1-1H Q11 | 2 marks
Estimate the median from a cumulative-frequency diagram

January 2021 | 4MA1-1H Q11 | 3 marks
Recover a threshold from a target count or amount

June 2021 | 4MA1-2H Q12 | 2 marks
Plot from an existing cumulative-frequency table

June 2021 | 4MA1-2H Q12 | 1 marks
Estimate the median from a cumulative-frequency diagram

June 2021 | 4MA1-2H Q12 | 2 marks
Estimate IQR from a cumulative-frequency diagram

... 7 more item(s) follow

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

13 The frequency table gives information about the times, in minutes, that 60 people took to complete a puzzle.

Time (t minutes)	**Frequency**
$0 < t \leqslant 10$	8
$10 < t \leqslant 20$	13
$20 < t \leqslant 30$	12
$30 < t \leqslant 40$	17
$40 < t \leqslant 50$	7
$50 < t \leqslant 60$	3

(a) Complete the cumulative frequency table.

Time (t minutes)	**Cumulative frequency**
$0 < t \leqslant 10$	
$0 < t \leqslant 20$	
$0 < t \leqslant 30$	
$0 < t \leqslant 40$	
$0 < t \leqslant 50$	
$0 < t \leqslant 60$	

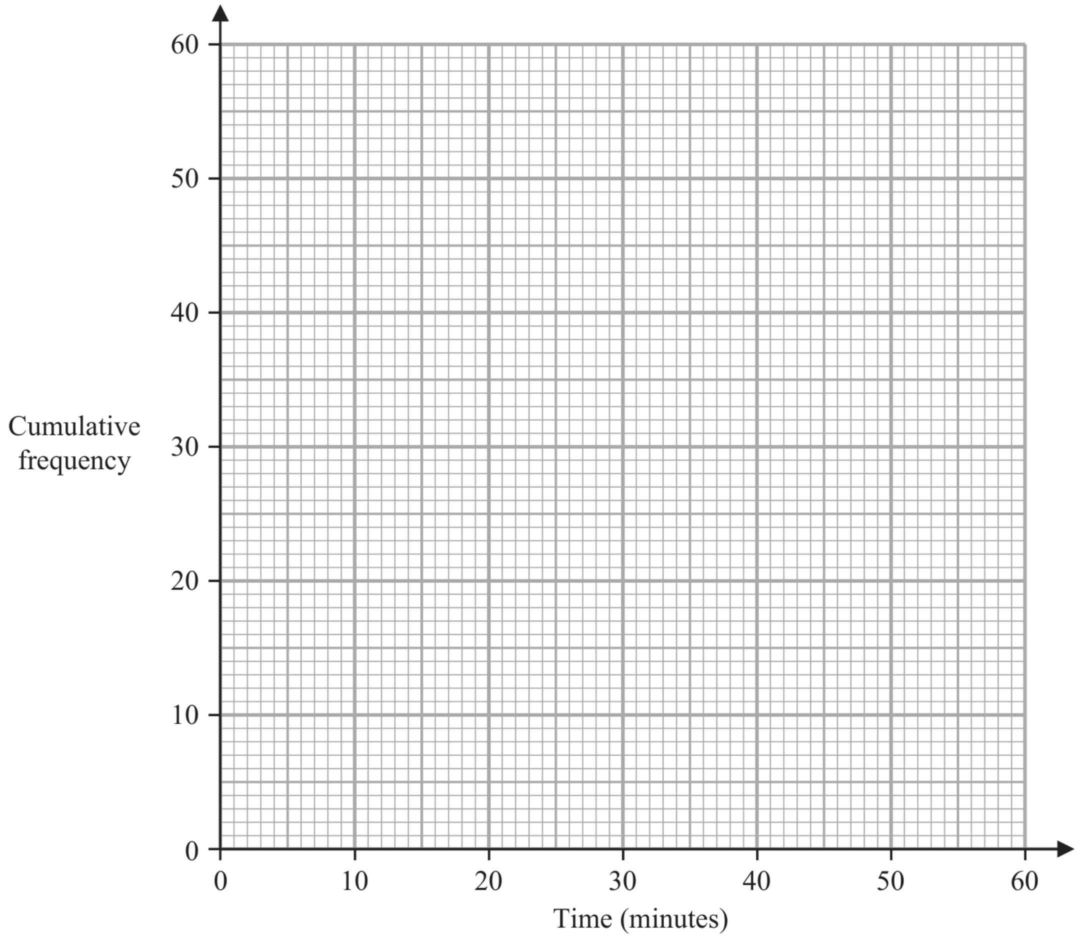
(1)

(b) On the grid opposite, draw a cumulative frequency graph for your table.

Continued

4MA1-2H Q13

UNIT 2



(2)

(c) Use your graph to find an estimate for the median time.

.... minutes

(1)

(d) Use your graph to find an estimate for the number of these people who took more than 45 minutes to complete the puzzle.

....

(2)

(Total for Question 13 is 6 marks)

Question 18
4MA1-2H Q13

MODULAR UNIT 2
7 marks

13 The table gives information about the distances, in km, that 70 teachers travel to school.

Distance (d km)	**Frequency**
$0 < d \leqslant 10$	7
$10 < d \leqslant 20$	17
$20 < d \leqslant 30$	18
$30 < d \leqslant 40$	14
$40 < d \leqslant 50$	10
$50 < d \leqslant 60$	4

(a) Complete the cumulative frequency table.

Distance (d km)	**Cumulative frequency**
$0 < d \leqslant 10$	
$0 < d \leqslant 20$	
$0 < d \leqslant 30$	
$0 < d \leqslant 40$	
$0 < d \leqslant 50$	
$0 < d \leqslant 60$	

(1)

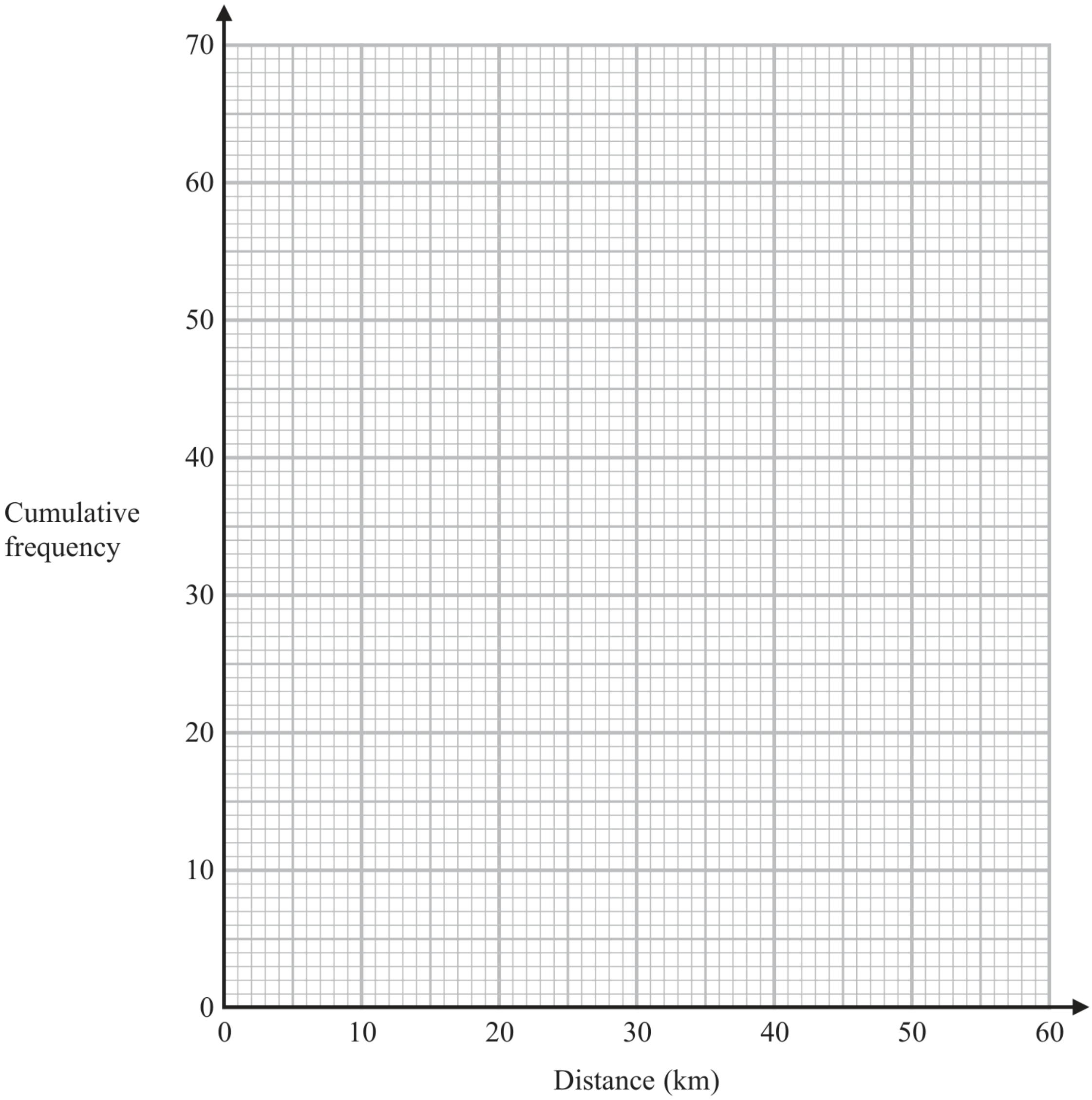
Question 18 (continued)

4MA1-2H Q13

MODULAR UNIT 2
7 marks

(b) On the grid opposite, draw a cumulative frequency graph for your table.

(2)



(c) Use your graph to find an estimate for the interquartile range of the distances.

.... km

(2)

Question 18 (continued)

4MA1-2H Q13

MODULAR UNIT 2
7 marks

(d) Use your graph to find an estimate for the number of teachers who travel more than 46 km.

....

(2)

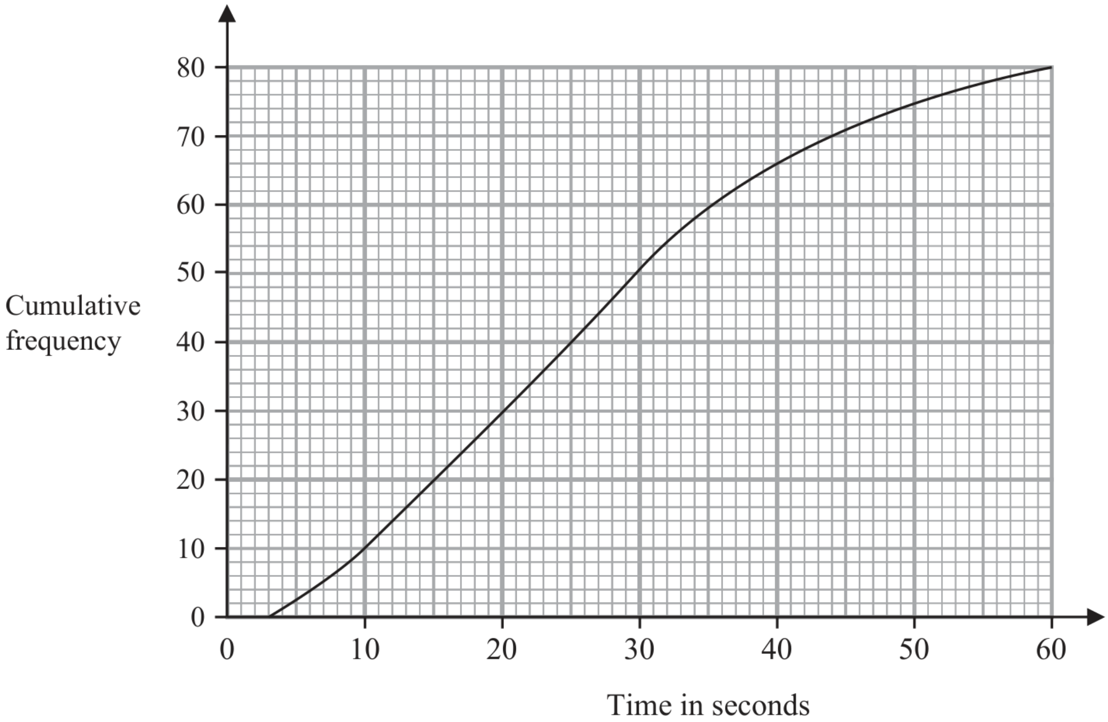
Question 09

4MA1-1H Q13 (a)

MODULAR UNIT 2

1 marks

13 The cumulative frequency graph gives information about the times, in seconds, that 80 adults took to log in to an online bank.



(a) Find an estimate for the median time.

.... seconds

(1)

WORKING SPACE

Working space for Question 09

Continue your method

UNIT 2
1 marks

WORKING SPACE

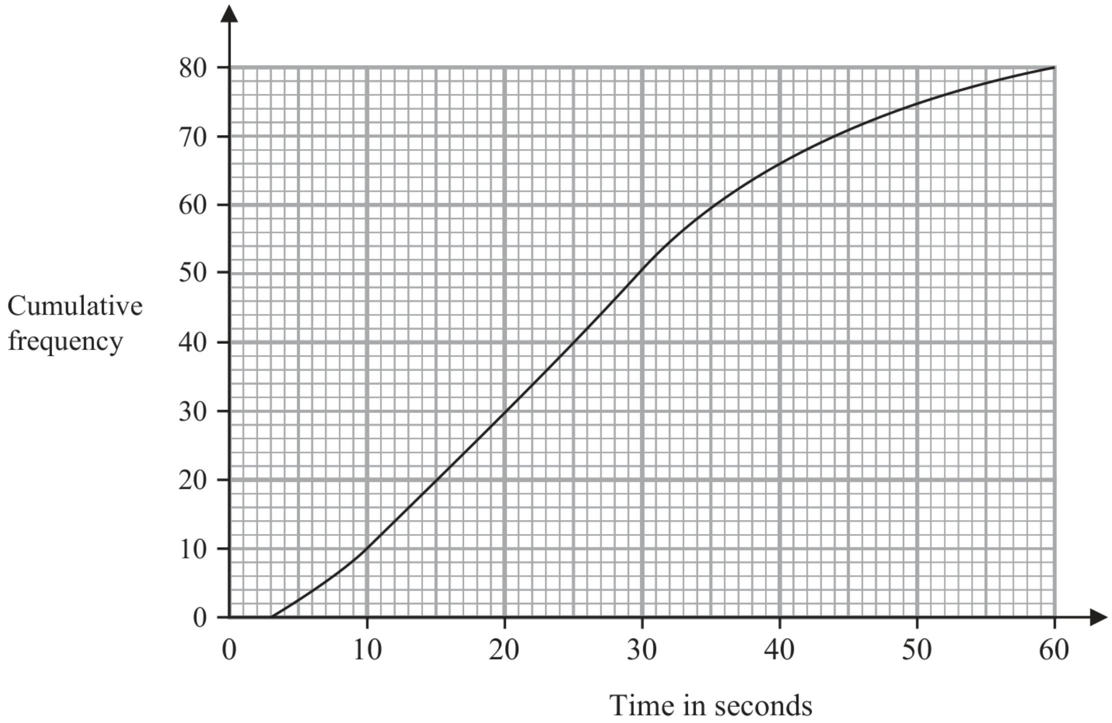
Question 10

4MA1-1H Q13 (b)

MODULAR UNIT 2

3 marks

13 The cumulative frequency graph gives information about the times, in seconds, that 80 adults took to log in to an online bank.



(b) Work out the percentage of these adults that took longer than 50 seconds to log in.
Show your working clearly.

....%

(3)

WORKING SPACE

Working space for Question 10

Continue your method

UNIT 2
3 marks

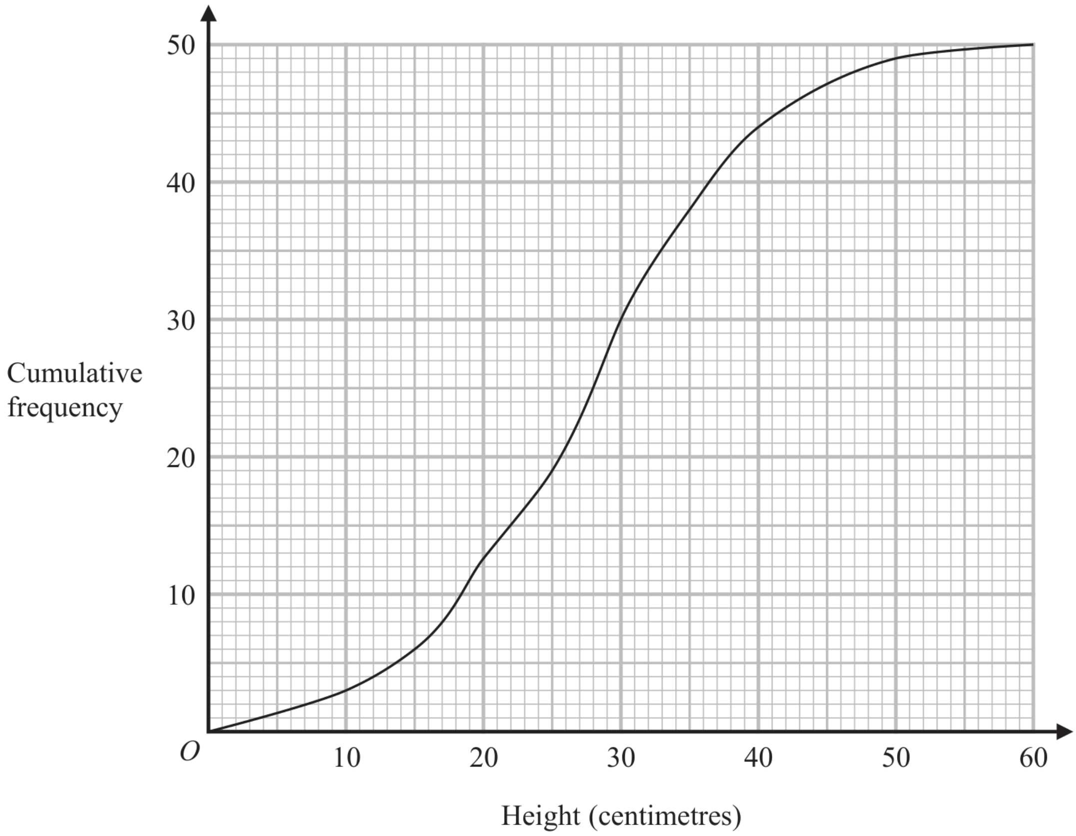
WORKING SPACE

Question 12

4MA1-1H Q12

MODULAR UNIT 2
4 marks

12 The cumulative frequency graph shows information about the heights, in centimetres, of 50 plants in a flowerbed.



(a) Use the graph to find an estimate for the median height of these plants.

.... centimetres
(1)

(b) Use the graph to find the frequency for the class interval $30 < \text{Height} \leqslant 40$

....
(1)

(c) Use the graph to find an estimate for the number of plants with a height greater than 35 centimetres.

....
(2)

Working space for Question 12
Continue your method

UNIT 2
4 marks

WORKING SPACE

Question 27

4MA1-2H Q13 M01, M02, M03, M04

MODULAR UNIT 2
6 marks

13 The frequency table gives information about the weights, in kilograms, of 60 parcels in a delivery van.

Weight (w kilograms)	Frequency
$0 < w \leqslant 1$	4
$1 < w \leqslant 2$	15
$2 < w \leqslant 3$	20
$3 < w \leqslant 4$	11
$4 < w \leqslant 5$	6
$5 < w \leqslant 6$	4

(a) Complete the cumulative frequency table.

Weight (w kilograms)	Cumulative frequency
$0 < w \leqslant 1$	
$0 < w \leqslant 2$	
$0 < w \leqslant 3$	
$0 < w \leqslant 4$	
$0 < w \leqslant 5$	
$0 < w \leqslant 6$	

(1)

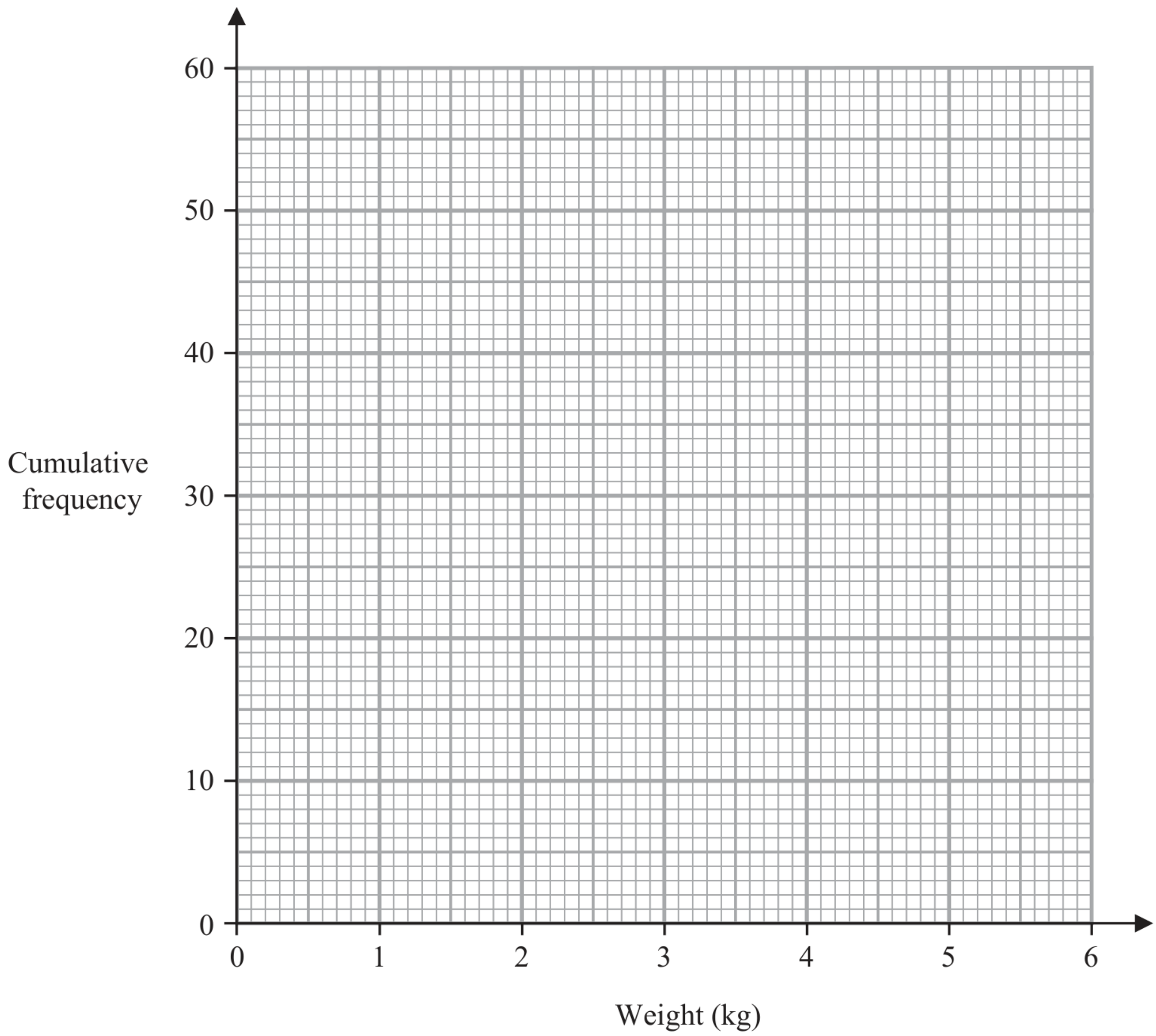
(b) On the grid opposite, draw a cumulative frequency graph for your table.

(2)

Question 27 (continued)

4MA1-2H Q13 M01, M02, M03, M04

MODULAR UNIT 2
6 marks



(c) Use your graph to find an estimate for the median weight of the 60 parcels.

.... kilograms
(1)

(d) Use your graph to find an estimate for the number of these parcels that weigh more than 3.7 kilograms.

....
(2)

Question 26

4MA1-2H Q13 Q13(a), Q13(b), Q13(c), Q13(d)

MODULAR UNIT 2
7 marks

13 The table gives information about the ages, in years, of 80 people in a train carriage.

Age (*a* years)	**Frequency**
$0 < a \leqslant 20$	7
$20 < a \leqslant 30$	25
$30 < a \leqslant 40$	20
$40 < a \leqslant 50$	14
$50 < a \leqslant 60$	8
$60 < a \leqslant 70$	6

(a) Complete the cumulative frequency table.

Age (*a* years)	**Cumulative frequency**
$0 < a \leqslant 20$	
$0 < a \leqslant 30$	
$0 < a \leqslant 40$	
$0 < a \leqslant 50$	
$0 < a \leqslant 60$	
$0 < a \leqslant 70$	

(1)

(b) On the grid opposite, draw a cumulative frequency graph for your table.

Question 26 (continued)

4MA1-2H Q13 Q13(a), Q13(b), Q13(c), Q13(d)

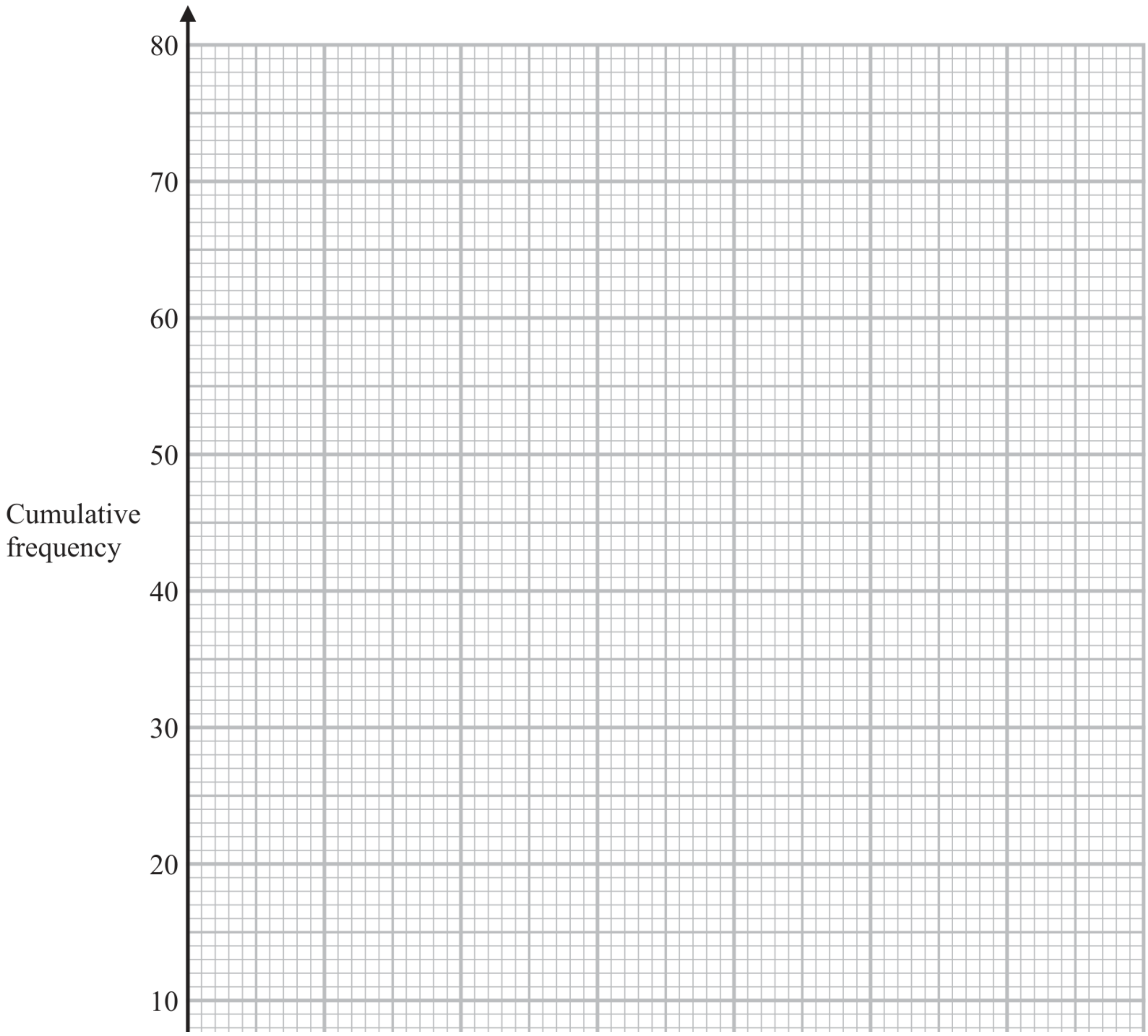
MODULAR UNIT 2
7 marks

(2)

(c) Use your graph to find an estimate for the median age of the 80 people.

.... years

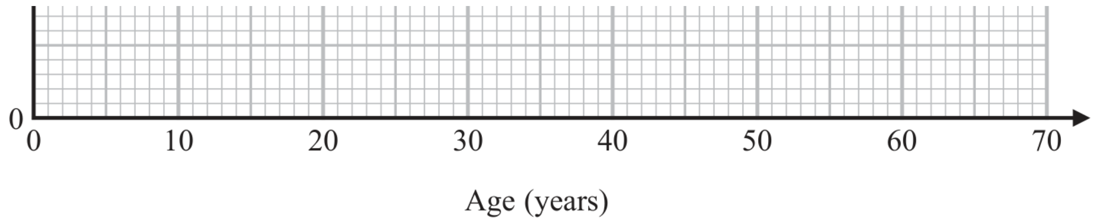
(1)



Question 26 (continued)

4MA1-2H Q13 Q13(a), Q13(b), Q13(c), Q13(d)

MODULAR UNIT 2
7 marks



Of the people in the train carriage, 60% of those who are aged between 18 and 65 are going to work. None of the other people in the train carriage are going to work.

(d) Use your graph to find an estimate for the number of people in the train carriage who are going to work.

....

(3)

Question 09

4MA1-1H Q12 Q12(a), Q12(b), Q12(c), Q12(d)

MODULAR UNIT 2
6 marks

12 The table gives information about the times, in minutes, taken by 80 customers to do their shopping in a supermarket.

Time taken (t minutes)	**Frequency**
$0 < t \leqslant 10$	7
$10 < t \leqslant 20$	26
$20 < t \leqslant 30$	24
$30 < t \leqslant 40$	14
$40 < t \leqslant 50$	7
$50 < t \leqslant 60$	2

(a) Complete the cumulative frequency table.

Time taken (t minutes)	**Cumulative frequency**
$0 < t \leqslant 10$	
$0 < t \leqslant 20$	
$0 < t \leqslant 30$	
$0 < t \leqslant 40$	
$0 < t \leqslant 50$	
$0 < t \leqslant 60$	

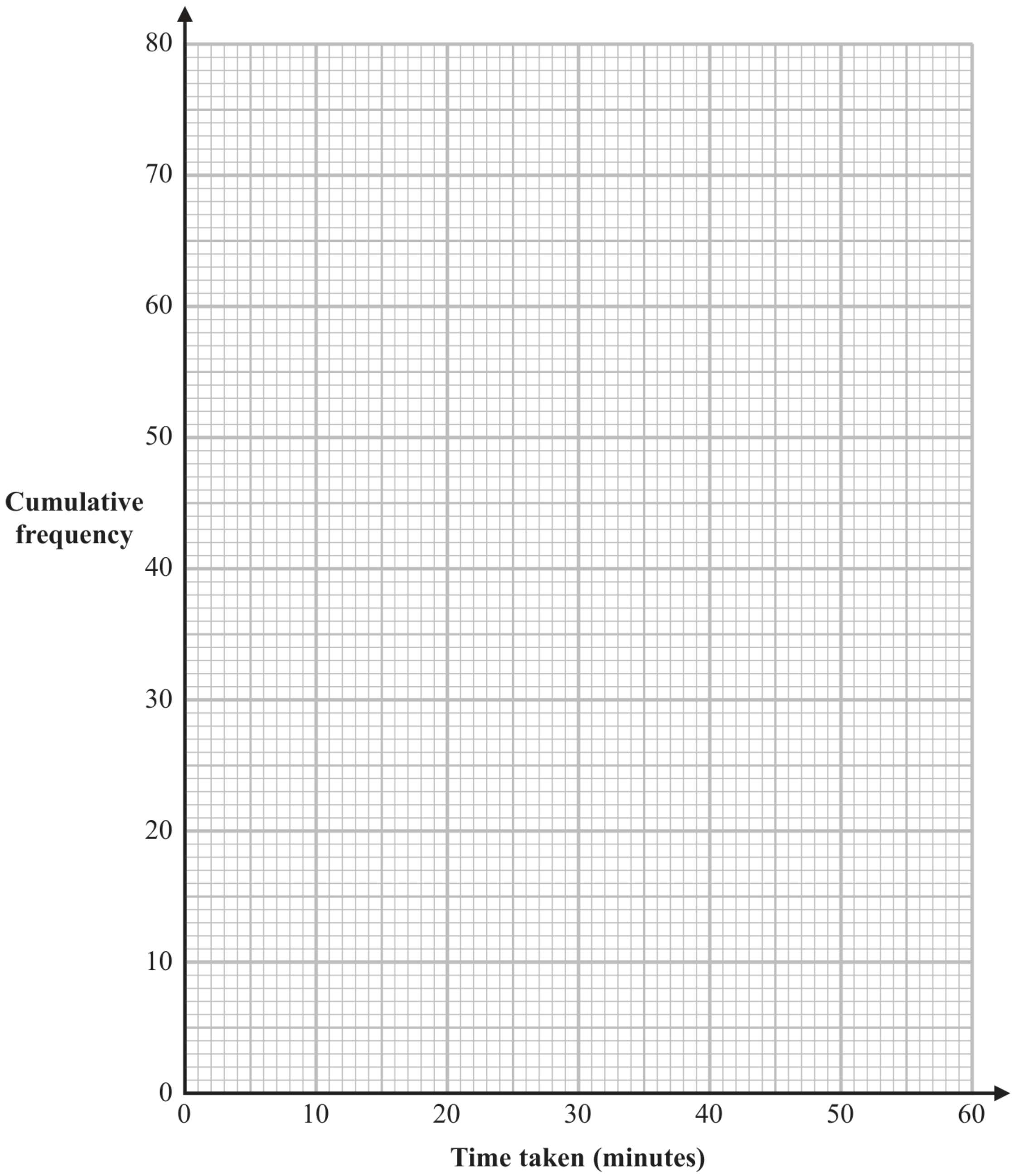
(1)

(b) On the grid opposite, draw a cumulative frequency graph for your table.

Question 09 (continued)

4MA1-1H Q12 Q12(a), Q12(b), Q12(c), Q12(d)

MODULAR UNIT 2
6 marks



(2)

Question 09 (continued)

4MA1-1H Q12 Q12(a), Q12(b), Q12(c), Q12(d)

MODULAR UNIT 2
6 marks

(c) Use your graph to find an estimate for the median time taken.

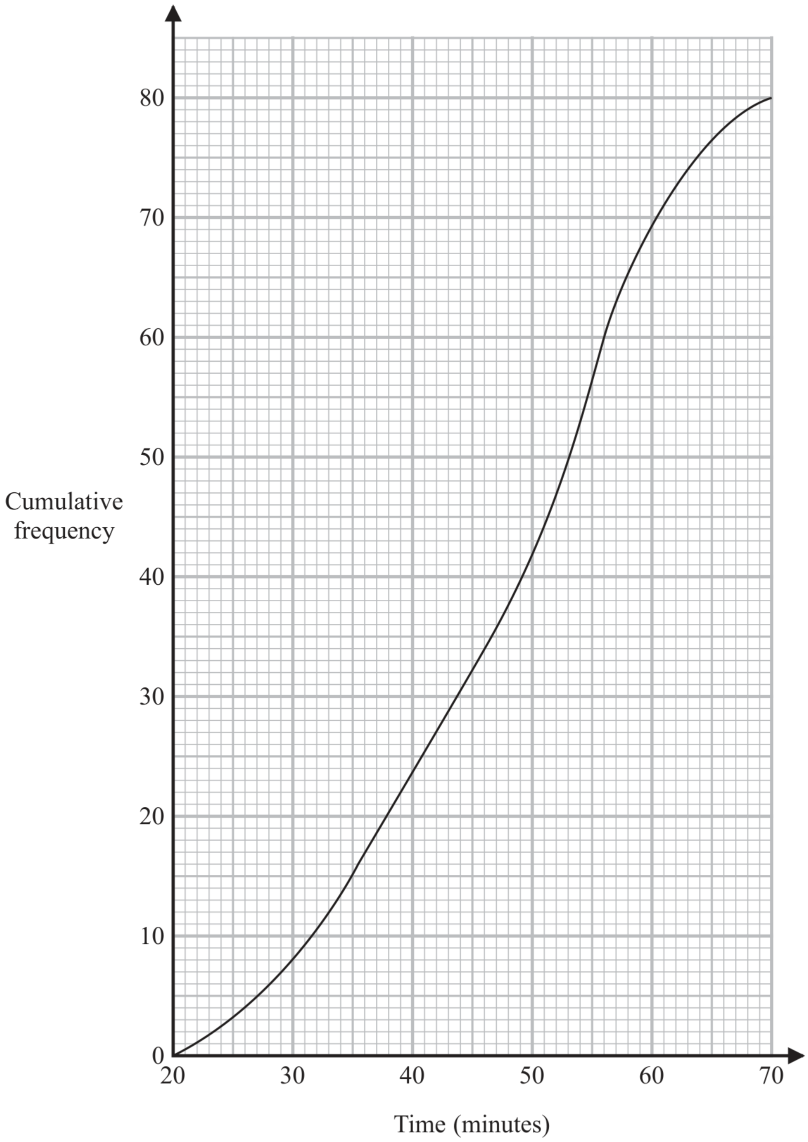
.... minutes
(1)

One of the 80 customers is chosen at random.

(d) Use your graph to find an estimate for the probability that the time taken by this customer was more than 42 minutes.

....
(2)

12 A total of 80 men and women took part in a race.
The cumulative frequency graph gives information about the times, in minutes, they took for the race.



(a) Use the graph to find an estimate for the interquartile range.

.... minutes

(2)

60% of the men took 50 minutes or less for the race.
No women took 50 minutes or less for the race.

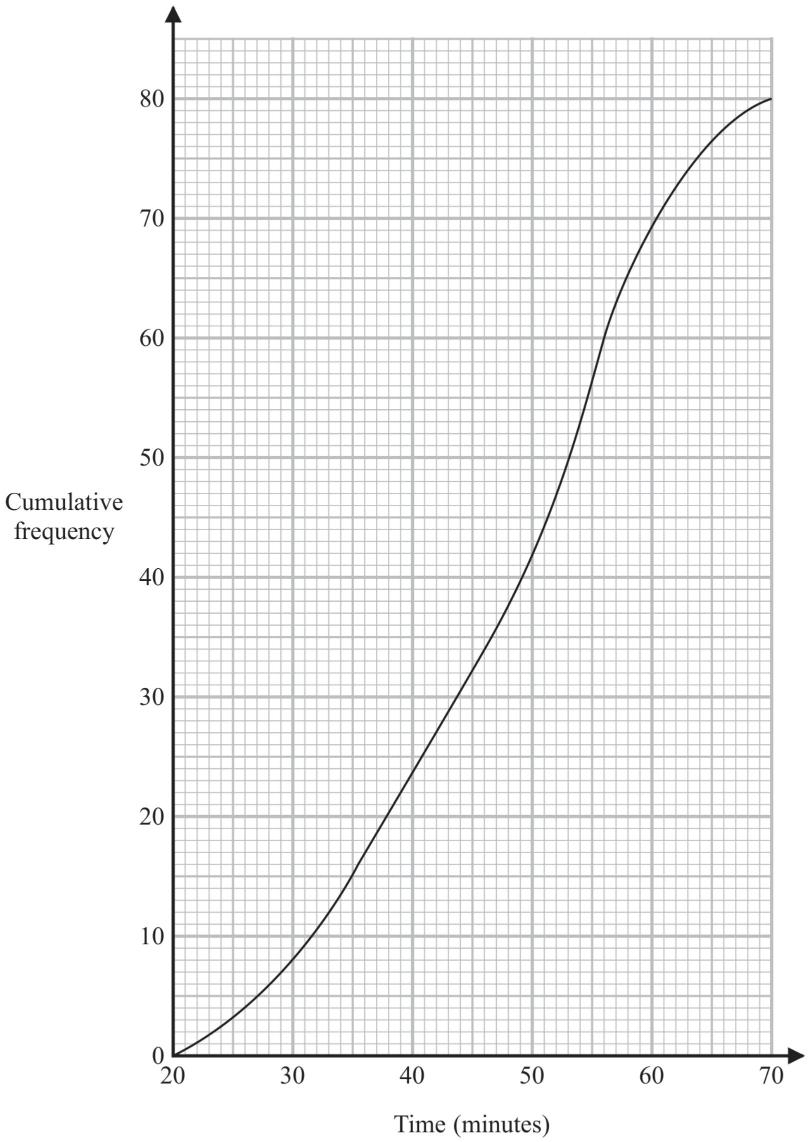
(b) Work out an estimate for the number of men who took part in the race.

....

(3)

YOUR WORKING

12 A total of 80 men and women took part in a race.
The cumulative frequency graph gives information about the times, in minutes, they took for the race.



(a) Use the graph to find an estimate for the interquartile range.

.... minutes

(2)

60% of the men took 50 minutes or less for the race.
No women took 50 minutes or less for the race.

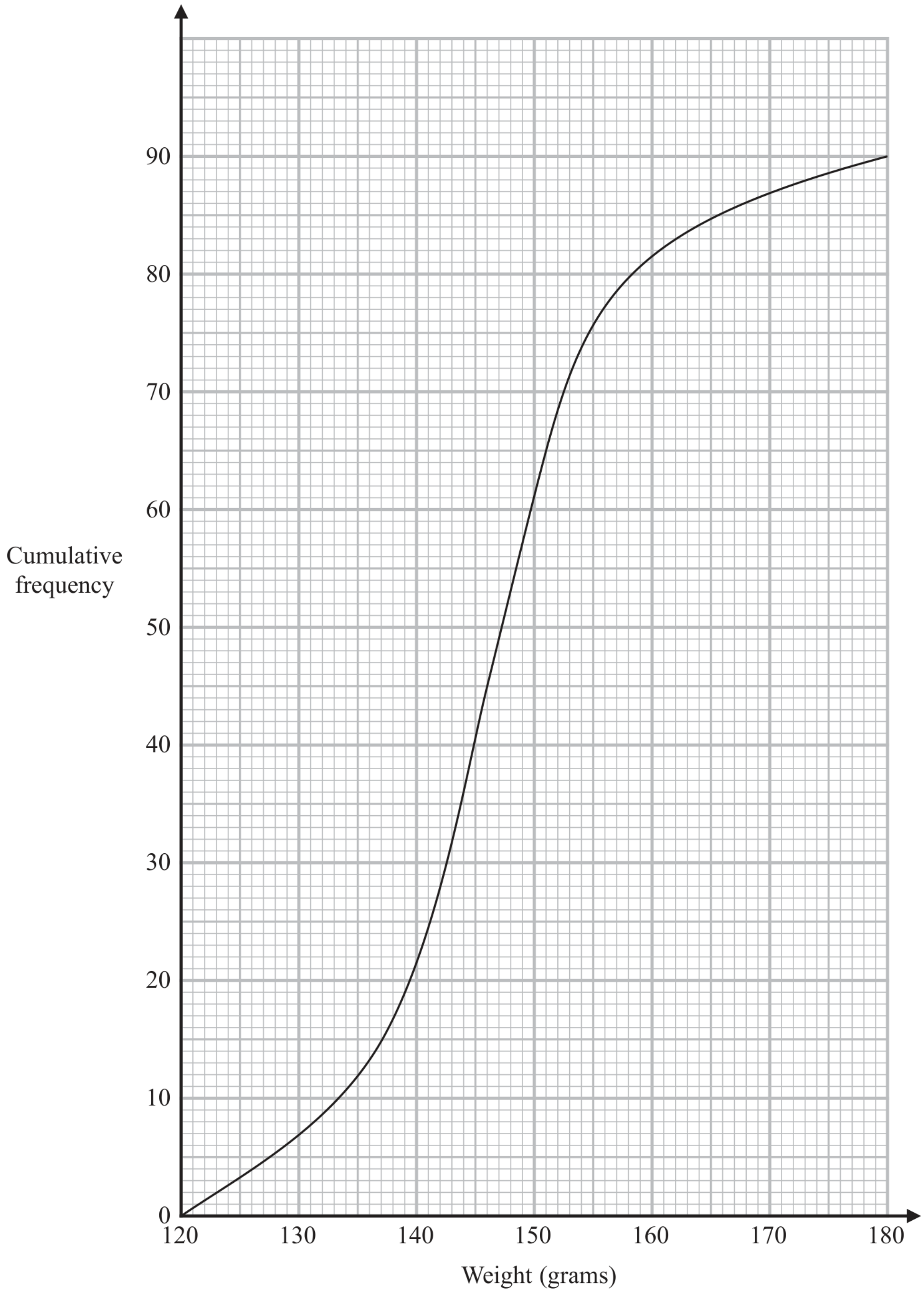
(b) Work out an estimate for the number of men who took part in the race.

....

(3)

YOUR WORKING

11 The cumulative frequency graph gives information about the weights, in grams, of 90 bags of sweets.



(a) Find an estimate for the median of the weights of these bags of sweets.

.... grams

(2)

YOUR WORKING

Roberto sells the bags of sweets to raise money for charity.
Bags with a weight greater than d grams are labelled large bags and sold for 3.75 euros each bag.

The total amount of money he receives by selling all the large bags is 93.75 euros.

(b) Find the value of d.

d = ...

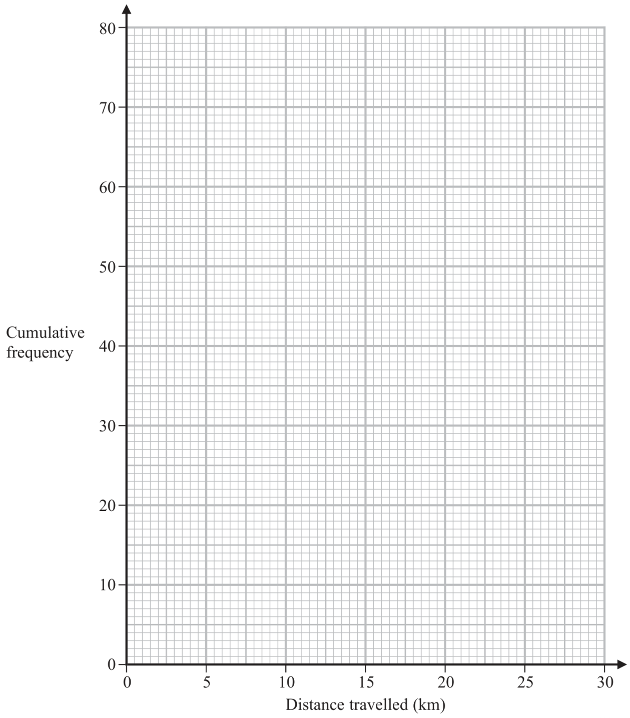
(3)

YOUR WORKING

12 The cumulative frequency table gives information about the distance, in kilometres, that each of 80 workers travel from home to work at Office A.

Distance travelled (d km)	Cumulative frequency
$0 < d \leqslant 5$	17
$0 < d \leqslant 10$	32
$0 < d \leqslant 15$	57
$0 < d \leqslant 20$	70
$0 < d \leqslant 25$	76
$0 < d \leqslant 30$	80

(a) On the grid below, draw a cumulative frequency graph for the information in the table.



(2)

(b) Use your graph to find an estimate for the median distance travelled.

.... km

(1)

(c) Use your graph to find an estimate for the interquartile range of the distances travelled.

.... km

(2)

For Office B, the median distance workers travel from home to work is 15 km and the interquartile range is 5 km.

(d) Use the information above to compare the distances that workers at Office A and workers at Office B travel from home to work.
Write down **two** comparisons.

1 ....

....

....

2 ....

....

....

(2)

YOUR WORKING

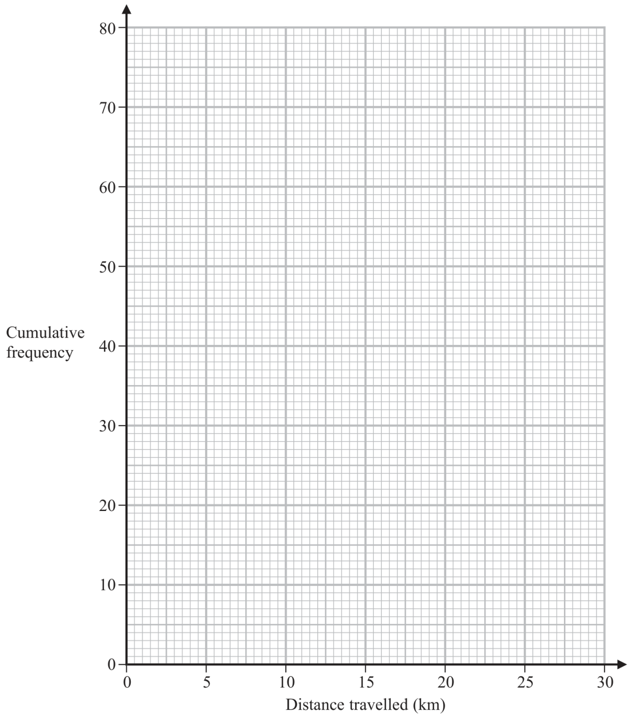
June 2021 | 4MA1-2H | Question 12 | 1 marks

Chapter OL-T55: Cumulative Frequency Diagrams

12 The cumulative frequency table gives information about the distance, in kilometres, that each of 80 workers travel from home to work at Office A.

Distance travelled (d km)	**Cumulative frequency**
$0 < d \leqslant 5$	17
$0 < d \leqslant 10$	32
$0 < d \leqslant 15$	57
$0 < d \leqslant 20$	70
$0 < d \leqslant 25$	76
$0 < d \leqslant 30$	80

(a) On the grid below, draw a cumulative frequency graph for the information in the table.



(2)

(b) Use your graph to find an estimate for the median distance travelled.

.... km

(1)

(c) Use your graph to find an estimate for the interquartile range of the distances travelled.

.... km

(2)

For Office B, the median distance workers travel from home to work is 15 km and the interquartile range is 5 km.

(d) Use the information above to compare the distances that workers at Office A and workers at Office B travel from home to work.
Write down **two** comparisons.

1 ....

....

....

2 ....

....

....

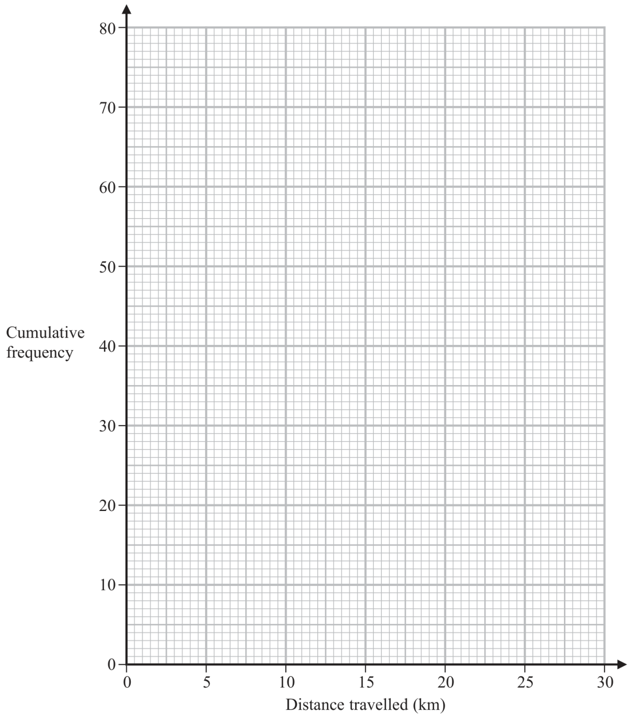
(2)

YOUR WORKING

12 The cumulative frequency table gives information about the distance, in kilometres, that each of 80 workers travel from home to work at Office A.

Distance travelled (d km)	**Cumulative frequency**
$0 < d \leqslant 5$	17
$0 < d \leqslant 10$	32
$0 < d \leqslant 15$	57
$0 < d \leqslant 20$	70
$0 < d \leqslant 25$	76
$0 < d \leqslant 30$	80

(a) On the grid below, draw a cumulative frequency graph for the information in the table.



(2)

(b) Use your graph to find an estimate for the median distance travelled.

.... km

(1)

(c) Use your graph to find an estimate for the interquartile range of the distances travelled.

.... km

(2)

For Office B, the median distance workers travel from home to work is 15 km and the interquartile range is 5 km.

(d) Use the information above to compare the distances that workers at Office A and workers at Office B travel from home to work.
Write down **two** comparisons.

1 ....

....

....

2 ....

....

....

(2)

YOUR WORKING

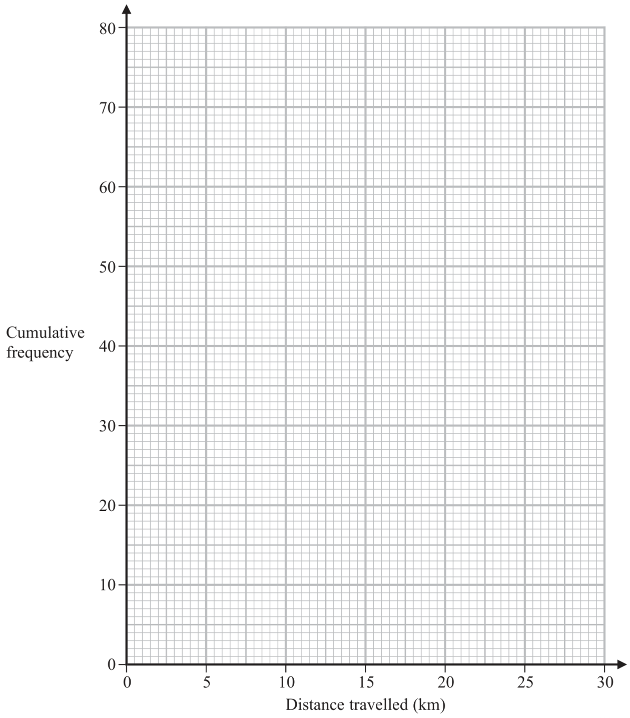
June 2021 | 4MA1-2H | Question 12 | 2 marks

Chapter OL-T55: Cumulative Frequency Diagrams

12 The cumulative frequency table gives information about the distance, in kilometres, that each of 80 workers travel from home to work at Office A.

Distance travelled (d km)	Cumulative frequency
$0 < d \leqslant 5$	17
$0 < d \leqslant 10$	32
$0 < d \leqslant 15$	57
$0 < d \leqslant 20$	70
$0 < d \leqslant 25$	76
$0 < d \leqslant 30$	80

(a) On the grid below, draw a cumulative frequency graph for the information in the table.



(2)

(b) Use your graph to find an estimate for the median distance travelled.

.... km

(1)

(c) Use your graph to find an estimate for the interquartile range of the distances travelled.

.... km

(2)

For Office B, the median distance workers travel from home to work is 15 km and the interquartile range is 5 km.

(d) Use the information above to compare the distances that workers at Office A and workers at Office B travel from home to work.
Write down **two** comparisons.

1 ....

....

....

2 ....

....

....

(2)

YOUR WORKING

11 The table gives information about the times taken by 90 runners to complete a 10 km race.

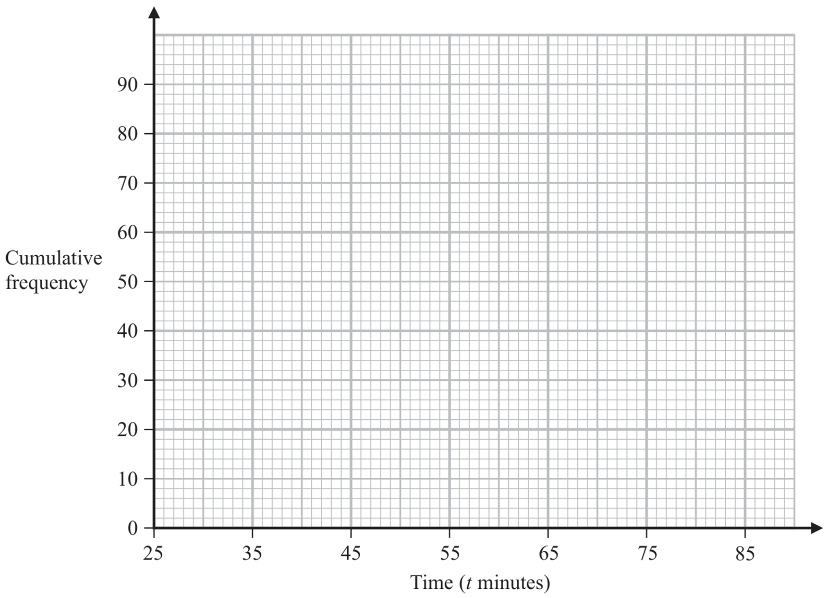
Time (t minutes)	Frequency
$25 < t \leqslant 35$	12
$35 < t \leqslant 45$	24
$45 < t \leqslant 55$	28
$55 < t \leqslant 65$	12
$65 < t \leqslant 75$	10
$75 < t \leqslant 85$	4

(a) Complete the cumulative frequency table.

Time (t minutes)	Cumulative frequency
$25 < t \leqslant 35$	12
$25 < t \leqslant 45$	
$25 < t \leqslant 55$	
$25 < t \leqslant 65$	
$25 < t \leqslant 75$	
$25 < t \leqslant 85$	

(1)

(b) On the grid below, draw a cumulative frequency graph for your table.



(2)

Any runner who completed the race in a time T minutes such that $42 < T \leqslant 52$ minutes was awarded a silver medal.

(c) Use your graph to find an estimate for the number of runners who were awarded a silver medal.

.... runners

(2)

YOUR WORKING

November 2021 | 4MA1-1H | Question 11 | 2 marks

Chapter OL-T55: Cumulative Frequency Diagrams

11 The table gives information about the times taken by 90 runners to complete a 10 km race.

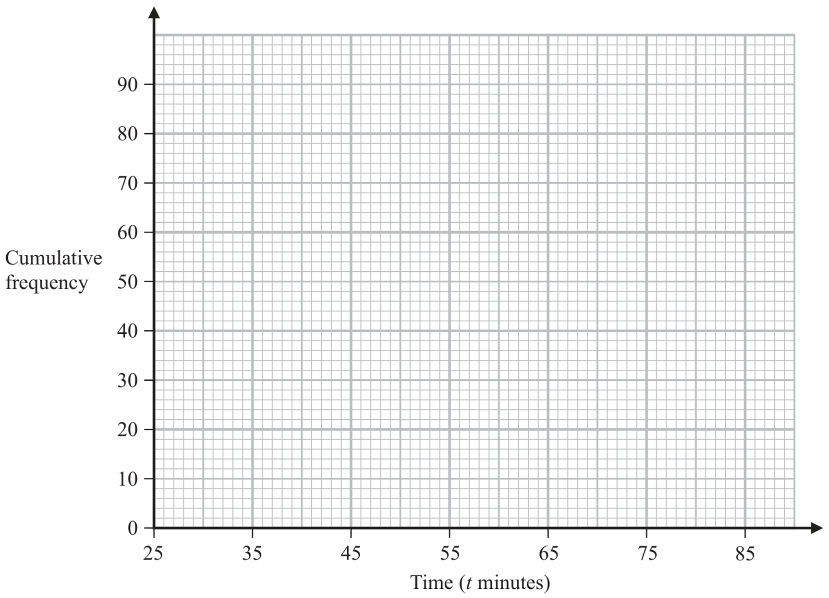
Time (t minutes)	Frequency
$25 < t \leqslant 35$	12
$35 < t \leqslant 45$	24
$45 < t \leqslant 55$	28
$55 < t \leqslant 65$	12
$65 < t \leqslant 75$	10
$75 < t \leqslant 85$	4

(a) Complete the cumulative frequency table.

Time (t minutes)	Cumulative frequency
$25 < t \leqslant 35$	12
$25 < t \leqslant 45$	
$25 < t \leqslant 55$	
$25 < t \leqslant 65$	
$25 < t \leqslant 75$	
$25 < t \leqslant 85$	

(1)

(b) On the grid below, draw a cumulative frequency graph for your table.



(2)

Any runner who completed the race in a time T minutes such that $42 < T \leqslant 52$ minutes was awarded a silver medal.

(c) Use your graph to find an estimate for the number of runners who were awarded a silver medal.

.... runners

(2)

YOUR WORKING

11 The table gives information about the times taken by 90 runners to complete a 10 km race.

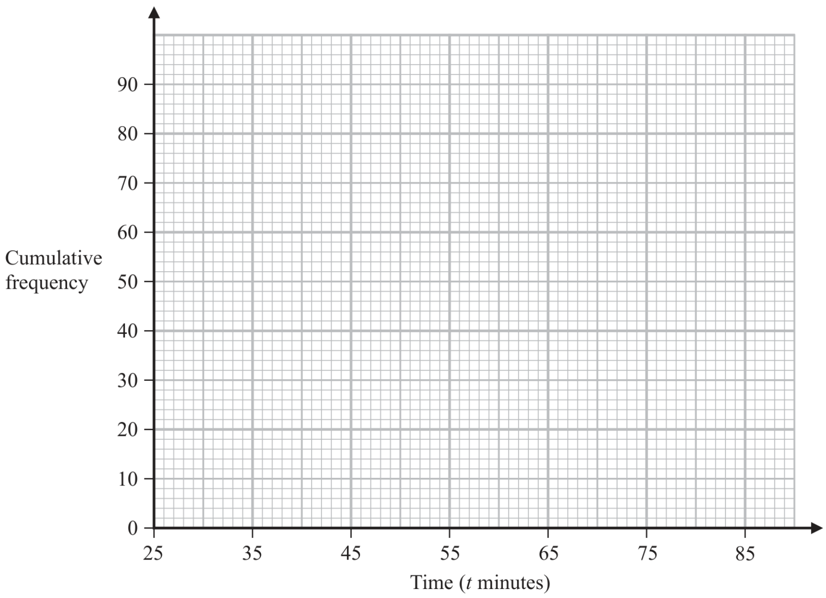
Time (t minutes)	Frequency
$25 < t \leqslant 35$	12
$35 < t \leqslant 45$	24
$45 < t \leqslant 55$	28
$55 < t \leqslant 65$	12
$65 < t \leqslant 75$	10
$75 < t \leqslant 85$	4

(a) Complete the cumulative frequency table.

Time (t minutes)	Cumulative frequency
$25 < t \leqslant 35$	12
$25 < t \leqslant 45$	
$25 < t \leqslant 55$	
$25 < t \leqslant 65$	
$25 < t \leqslant 75$	
$25 < t \leqslant 85$	

(1)

(b) On the grid below, draw a cumulative frequency graph for your table.



(2)

Any runner who completed the race in a time T minutes such that $42 < T \leqslant 52$ minutes was awarded a silver medal.

(c) Use your graph to find an estimate for the number of runners who were awarded a silver medal.

.... runners

(2)

YOUR WORKING

10 The table shows information about the number of minutes each of 120 buses was late last Monday.

Number of minutes late (L)	**Frequency**
$0 < L \leqslant 10$	10
$10 < L \leqslant 20$	16
$20 < L \leqslant 30$	44
$30 < L \leqslant 40$	29
$40 < L \leqslant 50$	15
$50 < L \leqslant 60$	6

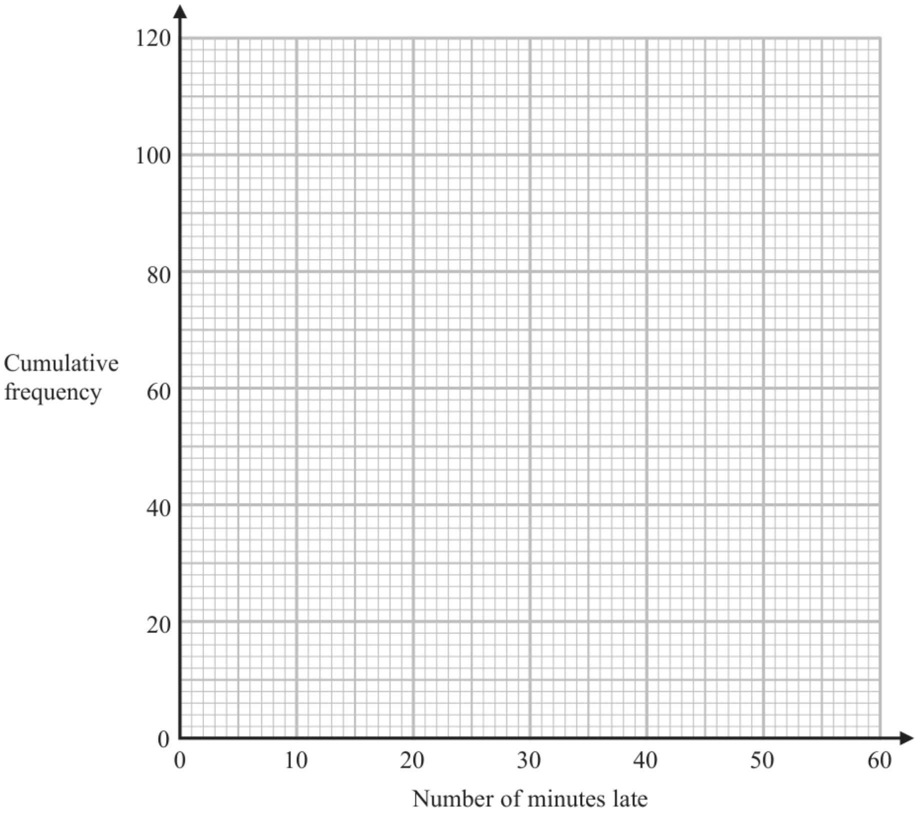
(a) Complete the cumulative frequency table below.

Number of minutes late (L)	**Cumulative frequency**
$0 < L \leqslant 10$	
$0 < L \leqslant 20$	
$0 < L \leqslant 30$	
$0 < L \leqslant 40$	
$0 < L \leqslant 50$	
$0 < L \leqslant 60$	

(1)

YOUR WORKING

(b) On the grid, draw a cumulative frequency graph for your table.



(2)

(c) Use your graph to find an estimate for the interquartile range.

.... minutes

(2)

(d) Use your graph to find an estimate for the number of buses that were more than 48 minutes late last Monday.

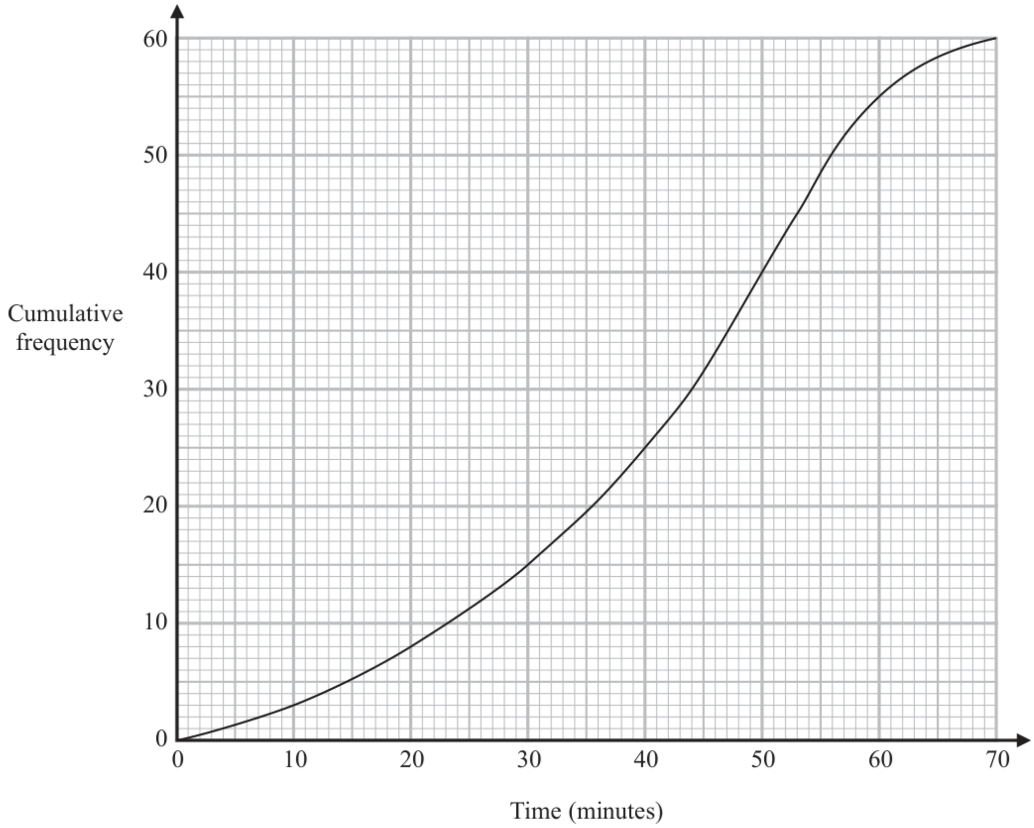
....

(2)

(Total for Question 10 is 7 marks)

YOUR WORKING

12 The cumulative frequency graph gives information about the time, in minutes, each of 60 people took to shop in a market.



(a) Use the graph to find an estimate for the median time people took to shop in the market.

.... minutes

(1)

(b) Use the graph to find an estimate for the number of people who took longer than 55 minutes to shop in the market.

....

(2)

YOUR WORKING

(c) Use the graph to complete the frequency table to give information about the time, in minutes, each of the 60 people took to shop in the market.

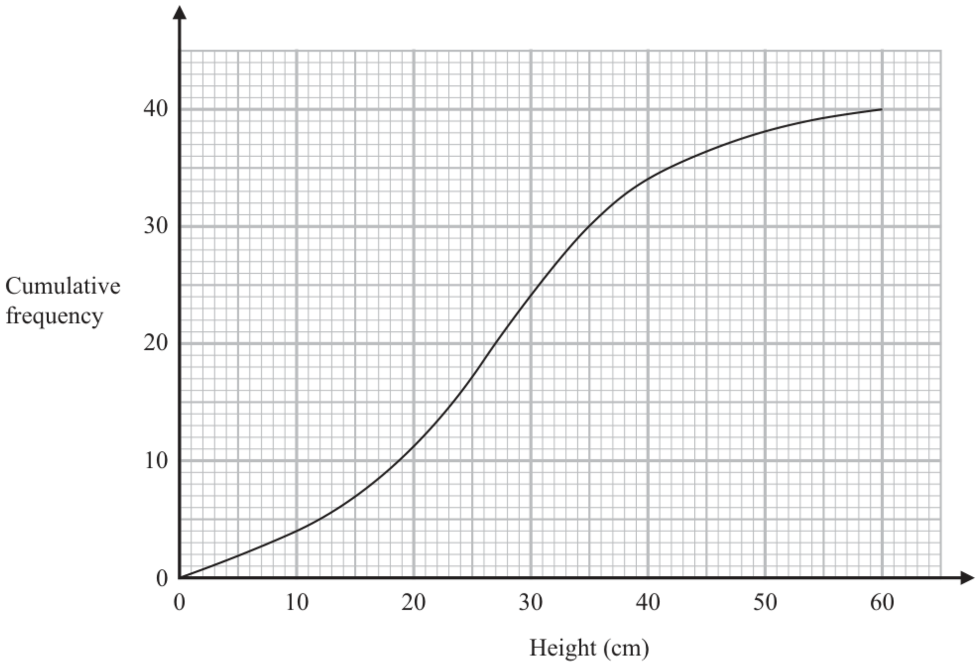
Time taken to shop in the market (m minutes)	**Frequency**
$0 < m \leqslant 10$	3
$10 < m \leqslant 20$	5
$20 < m \leqslant 30$	
$30 < m \leqslant 40$	
$40 < m \leqslant 50$	
$50 < m \leqslant 60$	
$60 < m \leqslant 70$	

(2)

(Total for Question 12 is 5 marks)

YOUR WORKING

14 The cumulative frequency graph shows information about the heights of 40 plants that Greta has grown.



(a) Use the graph to find an estimate for the median height.

.... cm

(1)

(b) Use the graph to find an estimate for the interquartile range of the heights.

.... cm

(2)

YOUR WORKING

Plants with a height greater than 40 cm are premium plants.
Greta sells all the premium plants for 30 euros each.

(c) Work out the total amount of money Greta receives for the premium plants.

.... euros

(2)

(Total for Question 14 is 5 marks)

YOUR WORKING